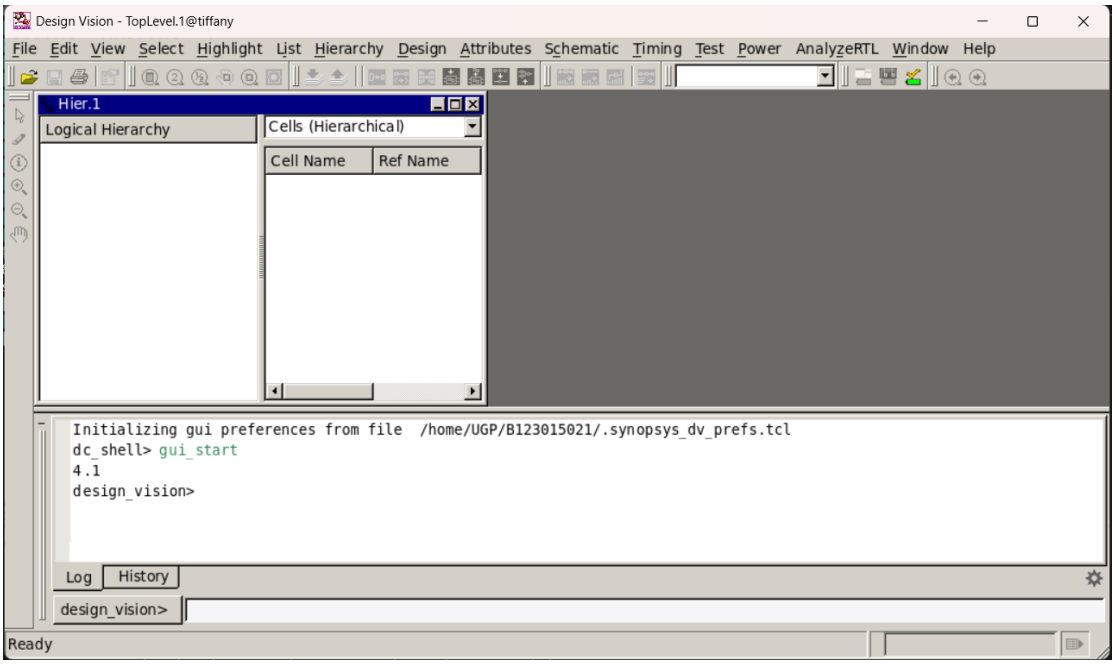
Practical Digital System Design Course - Logic Synthesis

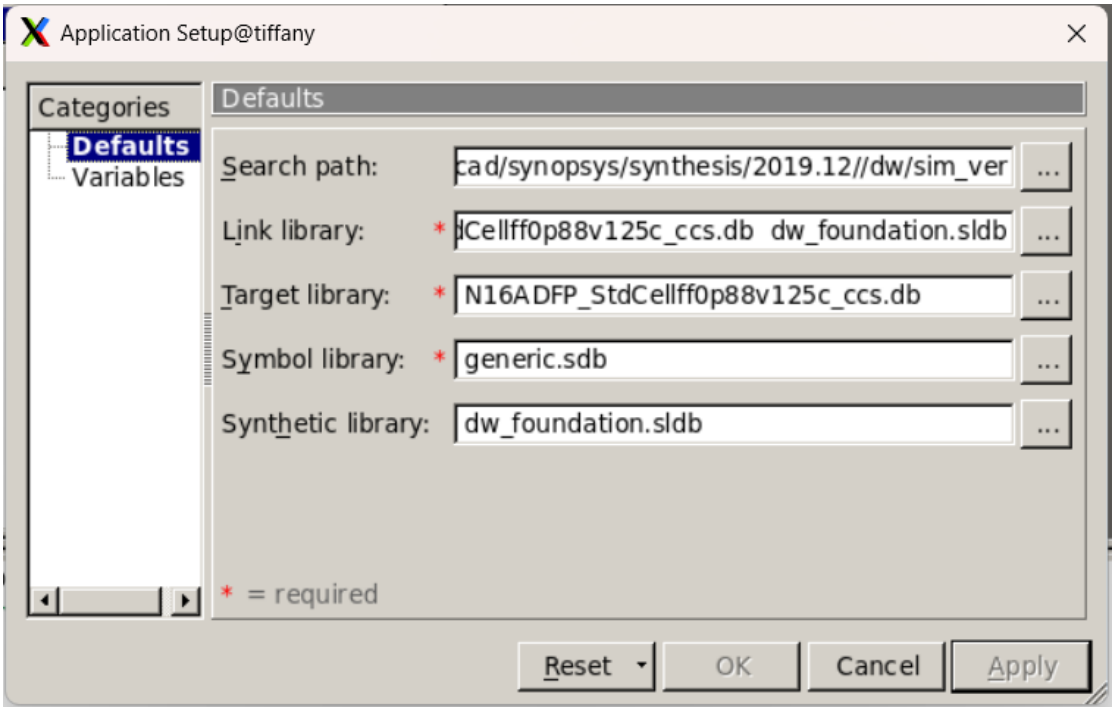
Implementation Worksheet (For In-Person Class Use)

Student ID: B123015021
Name: 尋以諾

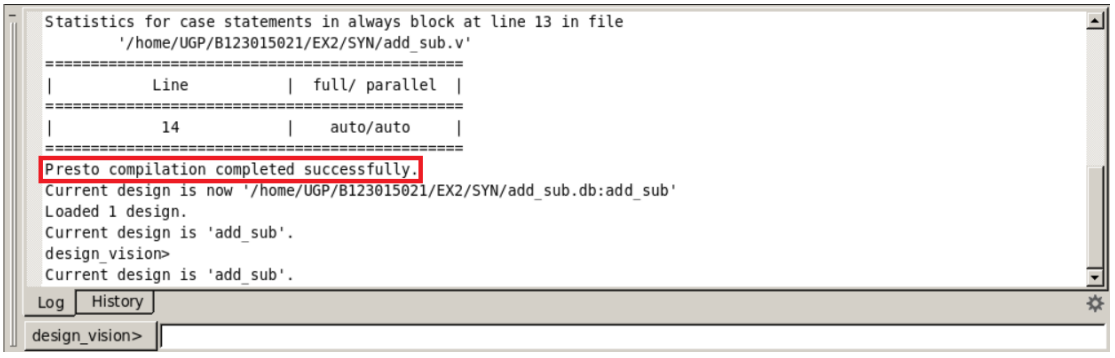
Based on the logic synthesis process introduced in class, please answer the following questions:

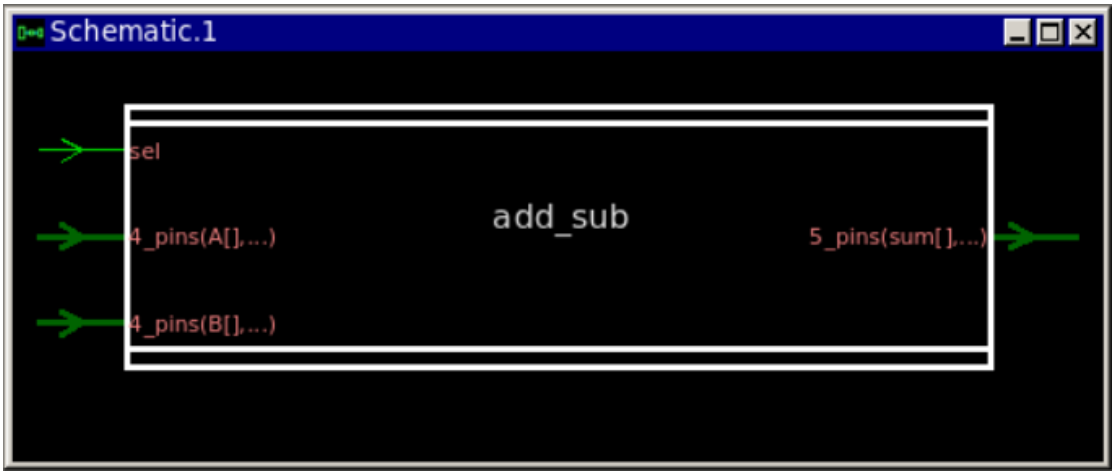
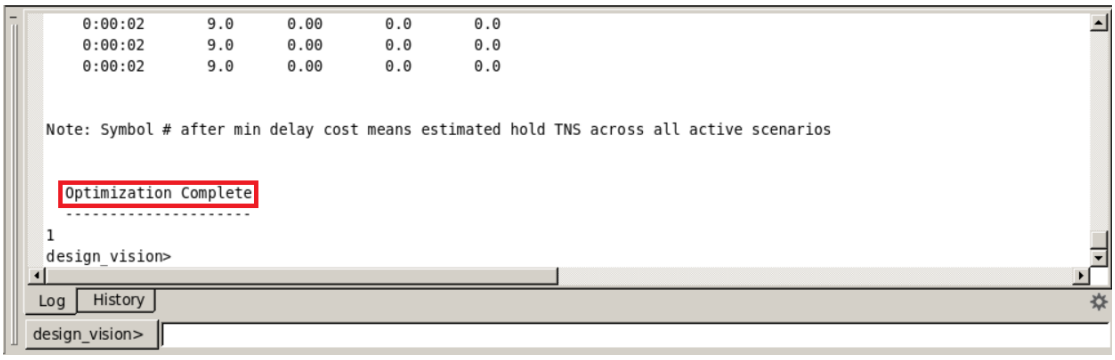
【Stage 1】 Setting Up and Launching Design Compiler

No.	Checklist
1	Please extract the EX2.zip file uploaded on the online university platform, upload it to the workstation, navigate to the EX2/SYN folder using Xshell, and confirm the presence of the Setup file by using the ls -a command. (Please attach a screenshot of the Xshell terminal and highlight the Setup file with a red box.) 
2	Ensure that Xlaunch is open and use the *dv &* command to launch Design Compiler. (Please attach a screenshot of the Design Compiler window.) 
3	According to Step 3 in Unit 5, confirm whether the Library has been loaded correctly.

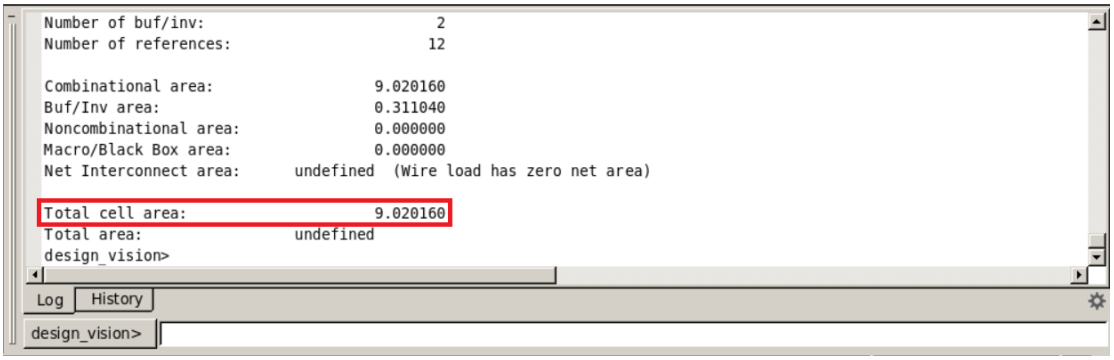
	(Please attach a screenshot of the Application Setup window.) 

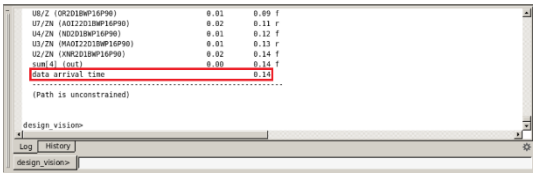
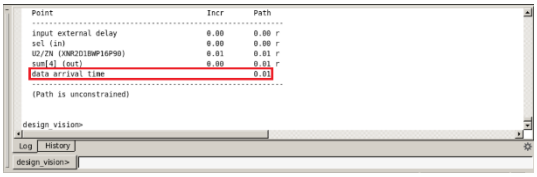
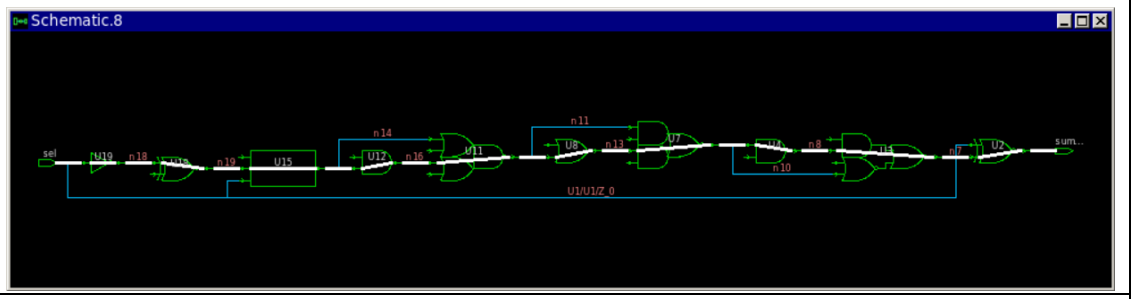
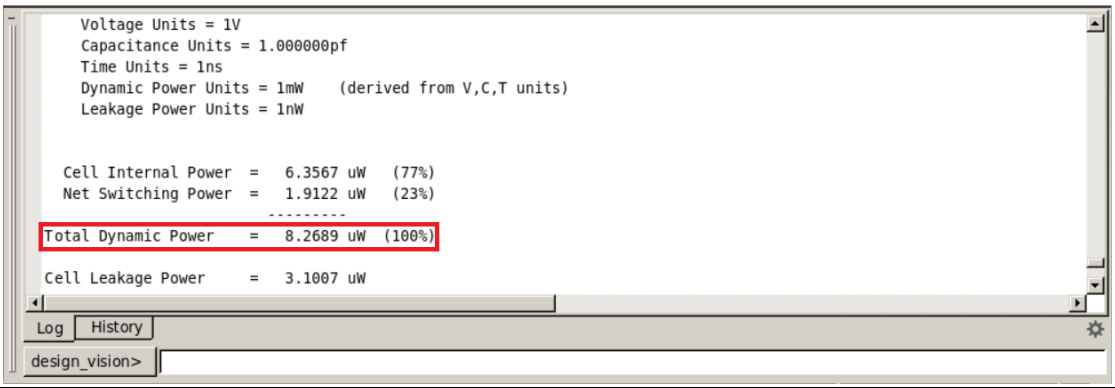
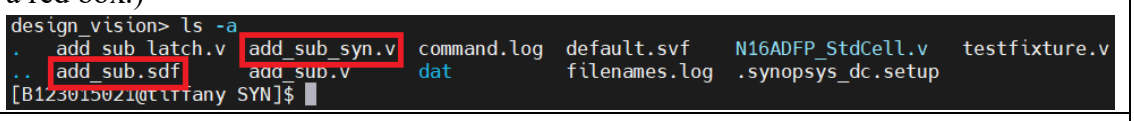
【Stage 2】: Logic Synthesis

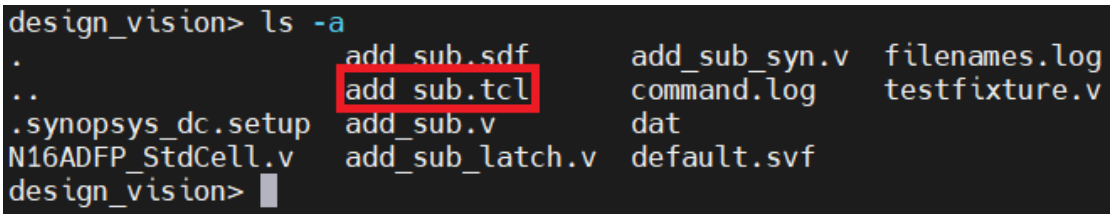
No.	Checklist
1	According to Step 4 in Unit 5, confirm that there are no error messages when reading the files.
	(Please attach a screenshot of the Design Compiler terminal and highlight the successful reading message with a red box.) 
2	According to Steps 5 and 6 in Unit 5, check if the internal circuit schematic is viewable.
	(Please attach a screenshot of the Schematic View window.)

	
3	According to Steps 7 and 9 in Unit 5, confirm that there are no error messages after synthesis. (Please attach a screenshot of the Design Compiler terminal and highlight the successful synthesis message with a red box.) 

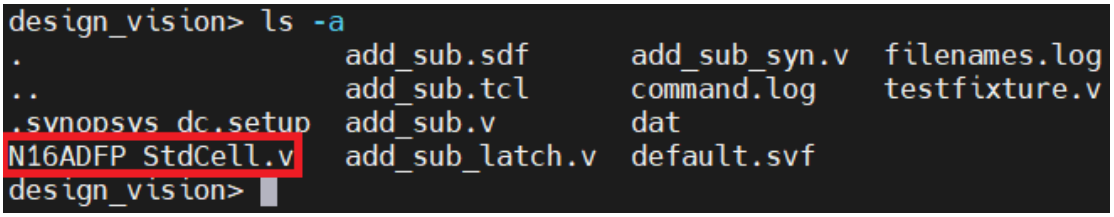
【Stage 3】: Performance Evaluation

No.	Checklist
1	According to Step 10 in Unit 5, confirm whether the Area Report is viewable. (Please attach a screenshot of the Area Report and highlight the Total cell area with a red box.) 
2	According to Step 11 in Unit 5, check if the Timing Report shows both max and min

<table>
<tr><td></td><td colspan="2">delay types.</td></tr>
<tr><td></td><td>(Please attach screenshots of the Timing Report for both max delay type, and highlight the Data arrival time with a red box.)
  </td><td>(Please attach screenshots of the Timing Report for both min delay type, and highlight the Data arrival time with a red box.)
  </td></tr>
<tr><td>3</td><td colspan="2">According to page 34 in Unit 5, check if it is possible to highlight or display the longest path separately.</td></tr>
<tr><td></td><td colspan="2">(Please attach a screenshot of the longest path.)
  </td></tr>
<tr><td>4</td><td colspan="2">According to Step 12 in Unit 5, confirm whether the Power Report is viewable.</td></tr>
<tr><td></td><td colspan="2">(Please attach a screenshot of the Power Report and highlight the Total Dynamic Power with a red box.)
  </td></tr>
<tr><td>5</td><td colspan="2">After Step 13 in Unit5, using Xshell to confirm with the ls -al command that the .v and .sdf files were saved successfully.</td></tr>
<tr><td></td><td colspan="2">(Please attach a screenshot of the Xshell terminal and highlight the .v and .sdf files with a red box.)
  </td></tr>
<tr><td>6</td><td colspan="2">In the Design Compiler terminal, select History, press Ctrl+A to select all commands, then click "Save Contents As..." and save it as add_sub.tcl. Then, using Xshell and the ls -al command, confirm that the .tcl file was saved successfully. Note: If you need to</td></tr>
</table>

	perform synthesis with the same settings in the future, you can open Design Compiler, type source add_sub.tcl in the terminal, and automatically execute the synthesis steps.
	(Please attach a screenshot of the Xshell terminal and highlight the .tcl file with a red box.) 

【Stage 4】: Post-Synthesis Simulation

No.	Cheklist
1	Using Xshell, check with the ls -al command if the N16ADFP_StdCell.v process file is in the current folder.
	(Please attach a screenshot of the Xshell terminal and highlight the process file with a red box.) 
2	According to Step 14 in Unit 5, confirm that the add_sub_syn .v file is imported, the N16ADFP_StdCell.v process file is included, and check if the timescale settings in the design file and TB file match.
	(Please attach a screenshot of the synthesized TB and highlight the modified parts of the TB with a red box.)

```
`timescale  10ns/100ps
//=======================================
//============= Include RTL ==============

// `include "./add_sub.v"

//============= Include SYN ==============

`include "./add_sub_syn.v"
`include "./N16ADFP_StdCell.v"

//=======================================

module testfixture;

//=======================================
//============= Define Path ==============

`define     PAT "./dat/PAT.dat"
`define     EXP "./dat/EXP.dat"

//========== Define Parameters ===========

`define     CYCLE   10
`define     PAT_NUM 5

//=======================================
//============ Declare Array ============

reg     [8:0]   pat_memory  [0:`PAT_NUM-1];
reg     [4:0]   exp_memory  [0:`PAT_NUM-1];

//============= Module Port ==============

reg             sel;
reg     [3:0]   A, B;

wire    [4:0]   sum;

//========== Varification variable ==========

integer         i;

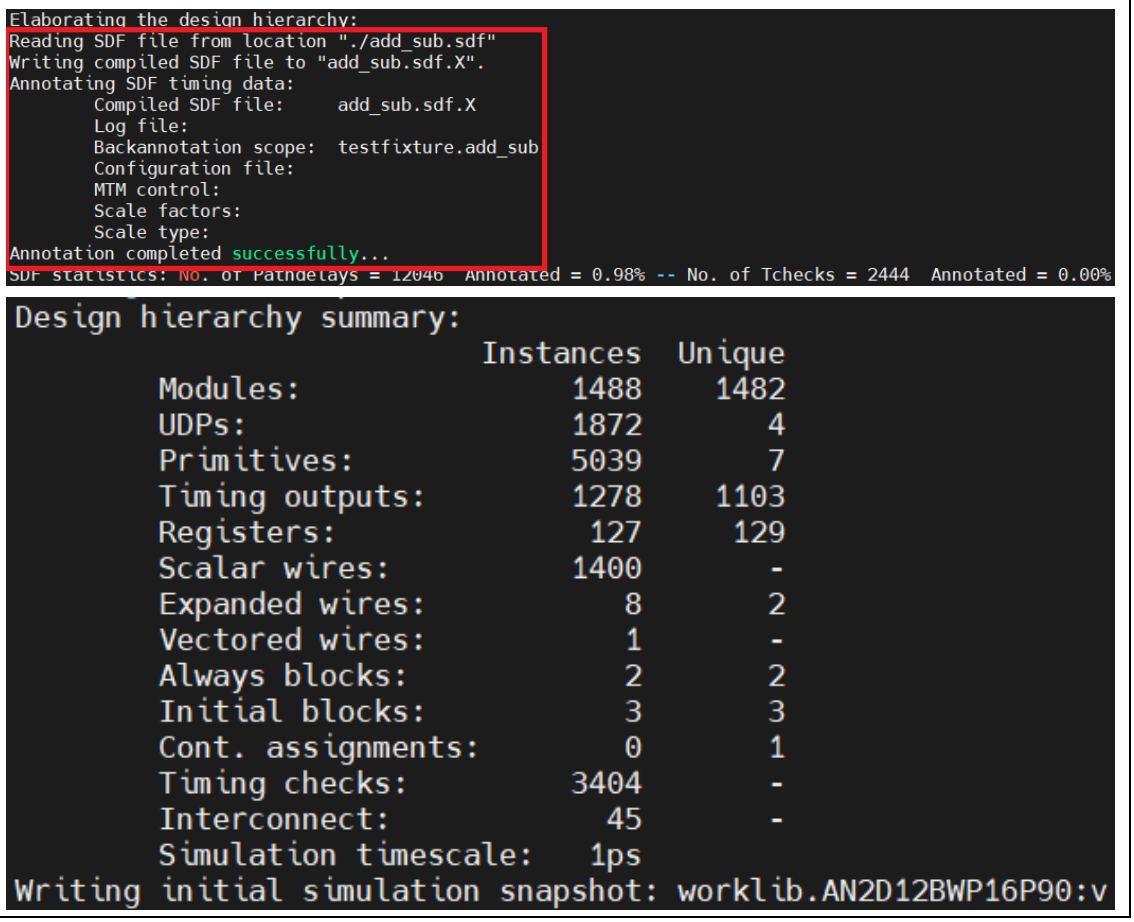
wire    [4:0]   result;

reg             clk, finish;
reg     [2:0]   pass_cnt, err_cnt;
reg     [4:0]   exp_tmp;

//=======================================
//================ SDF ==================

initial $sdf_annotate("./add_sub.sdf", add_sub);

//=======================================
```

3	Use the ncverilog testfixture.v +access+r command in the Xshell terminal to simulate the synthesized circuit and check for the "Annotation complete successfully" message along with the Totoro symbol. (Please attach a screenshot of the Xshell terminal and highlight the successful .sdf import message with a red box.) 

```
============================ Simulation START !! ============================


                     Pattern01:  ** Correct!! **
                     Pattern02:  ** Correct!! **
                     Pattern03:  ** Correct!! **
                     Pattern04:  ** Correct!! **
                     Pattern05:  ** Correct!! **


============================ Simulation Finish ! ============================


                ****************************
                **                        **       /|__/|
                **    Congratulations !!  **      / 0,0  |
                **                        **    /_____   |
                **  Simulation Complete!! **   /^ ^ ^ \  |
                **                        **  |^ ^ ^ ^ |w|
                ****************************   \m___m__|_|


Simulation complete via $finish(1) at time 57 NS + 0
./testfixture.v:145     #5 $finish;
```

【Stage 5】: Comparing Synthesis Results of Different Implementations

No.	Checklist
1	Synthesize add_sub_latch.v, compare the differences, and check if Latch messages appear in the Design Compiler terminal. (Please attach a screenshot of the Design Compiler terminal and highlight the Latch messages with a red box.)

```
Compiling source file /home/UGP/B123015021/EX2/SYN/add_sub_latch.v

Statistics for case statements in always block at line 15 in file
        '/home/UGP/B123015021/EX2/SYN/add_sub_latch.v'
===============================================
|           Line           |  full/ parallel  |
===============================================
|            16            |     no/auto      |
===============================================

Inferred memory devices in process
        in routine add_sub line 15 in file
                '/home/UGP/B123015021/EX2/SYN/add_sub_latch.v'.
=========================================================================
|    Register Name    | Type  | Width | Bus | MB | AR | AS | SR | SS | ST |
=========================================================================
|       sum_reg       | Latch |   5   |  Y  | N  | N  | N  | -  | -  | -  |
=========================================================================
Presto compilation completed successfully.
```

【File Description】: EX2.zip

Folder	File Name		Description
RTL	dat	EXP.dat	Output comparison data, used in the automated Testbench
		PAT.dat	Input test data, used in the automated Testbench
	add_sub.v		4-bit adder-subtractor circuit
	tb.v		Pre-synthesis automated Testbench
SYN	dat	EXP.dat	Output comparison data, used in the automated Testbench
		PAT.dat	Input test data, used in the automated Testbench
	.synopsys_dc.setup		Setup file (sets synthesis Library and paths)
	N16ADFP_StdCell.v		TSMC ADFP 16nm process file
	add_sub.v		4-bit adder-subtractor circuit
	add_sub_latch.v		4-bit adder-subtractor circuit (Latch version)
	testfixture.v		Post-synthesis automated Testbench

【**Note**】:

Each student must submit a PDF file to the EX2 folder in the assignment FTP by 11:59 AM on 2025/11/4 (Tue.).

Please Name the PDF file as EX2_yourID.
Do not zip the file